

# **Exercise: Filtering**

**Image Processing & Analysis for Life Scientists**

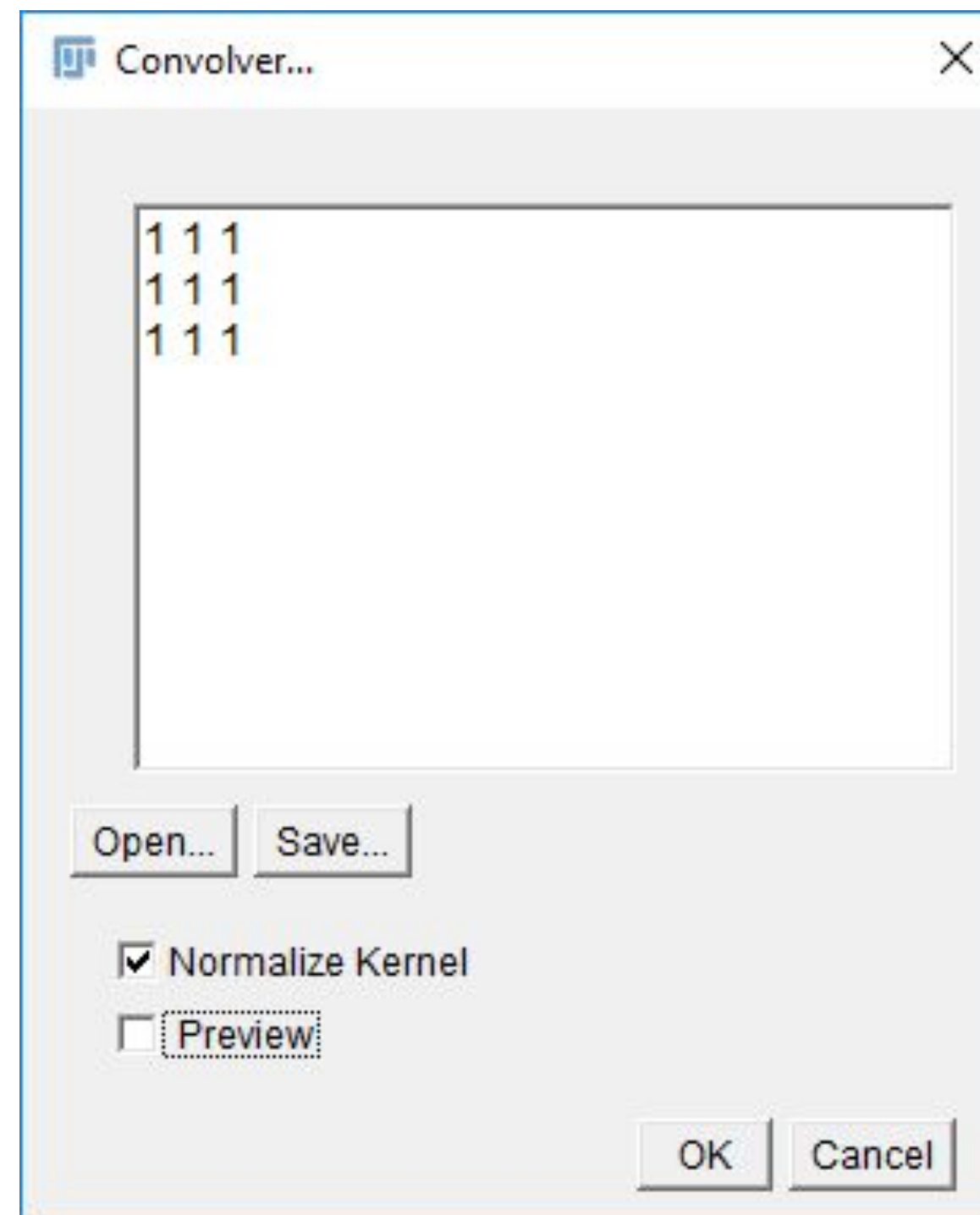
Olivier Burri, Romain Guiet & Arne Seitz

# Spatial Filtering: Convolution

1. Open *blobs.gif*
2. Zoom in till you see pixels
3. Launch  
***Process*→*Filter*→*Convolve***  
Match the kernel to the  
examples and use 'preview'

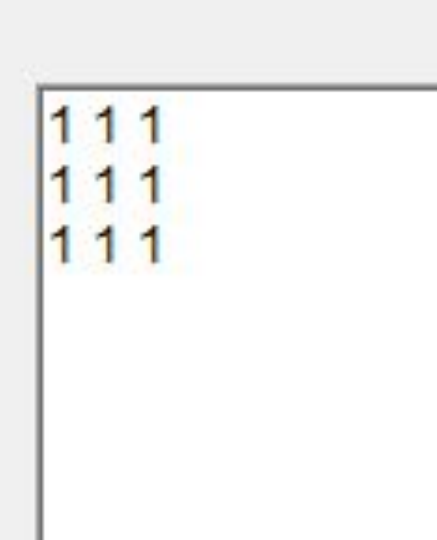
3x3

|   |   |   |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 1 | 1 |
| 1 | 1 | 1 |





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FÉDÉRALE DE LAUSANNE

- 
- Convolver...
- 1 1 1  
1 1 1  
1 1 1
- Open... Save...
- ☒ Normalize Kernel  
☐ Preview
- OK Cancel

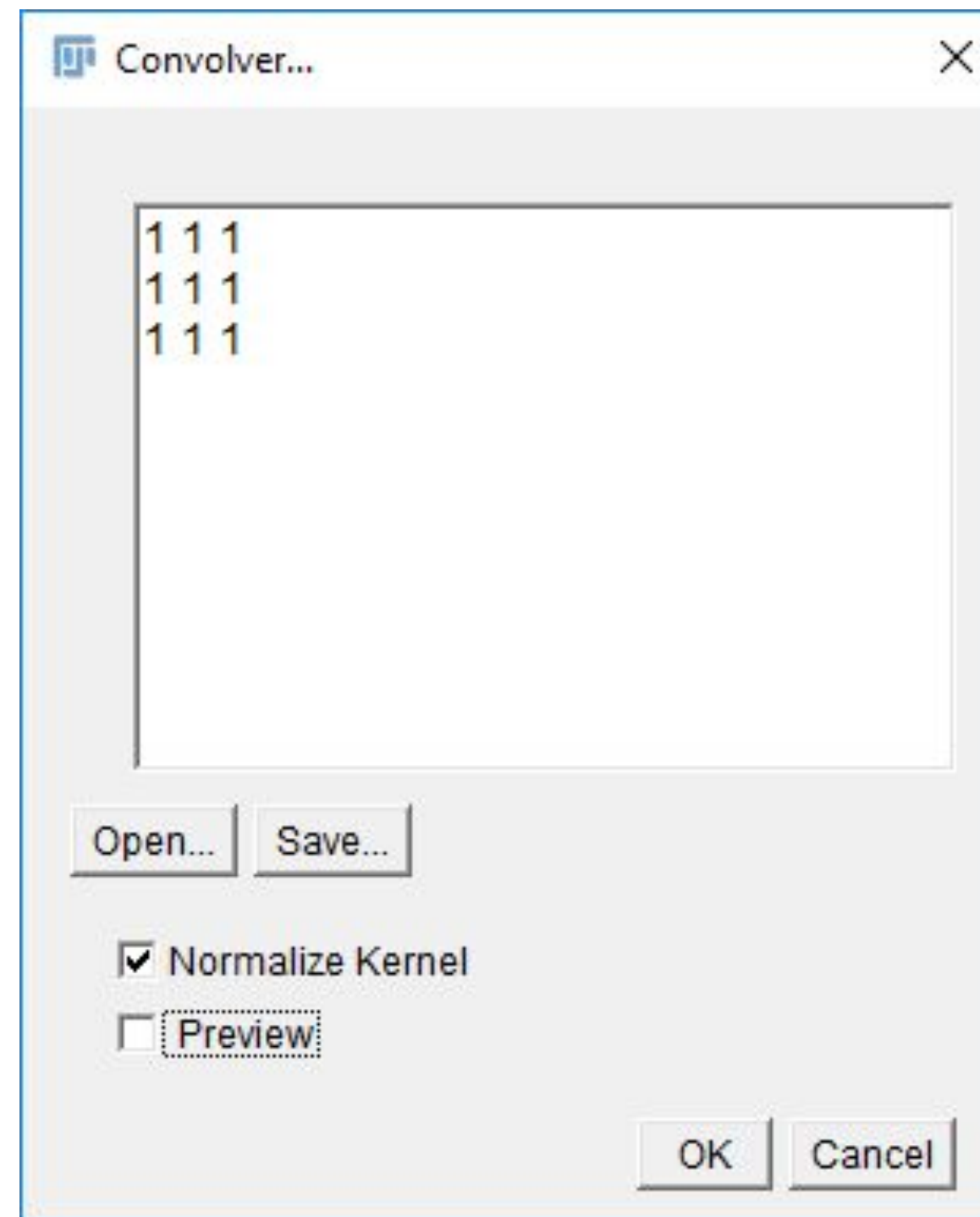
[illegible]





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1. Open *blobs.gif*  
Zoom in till you see pixels
2. Launch  
*Process*→*Filter*→*Convolve*  
Match the kernel to the examples and use 'preview'
3. What happens as the kernel gets larger?
4. What happens if you uncheck "Normalize Kernel"?
5. Try the **non-linear filter**  
'Median'  
*Process*→*Filter*→*Median*



3x3

|   |   |   |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 1 | 1 |
| 1 | 1 | 1 |

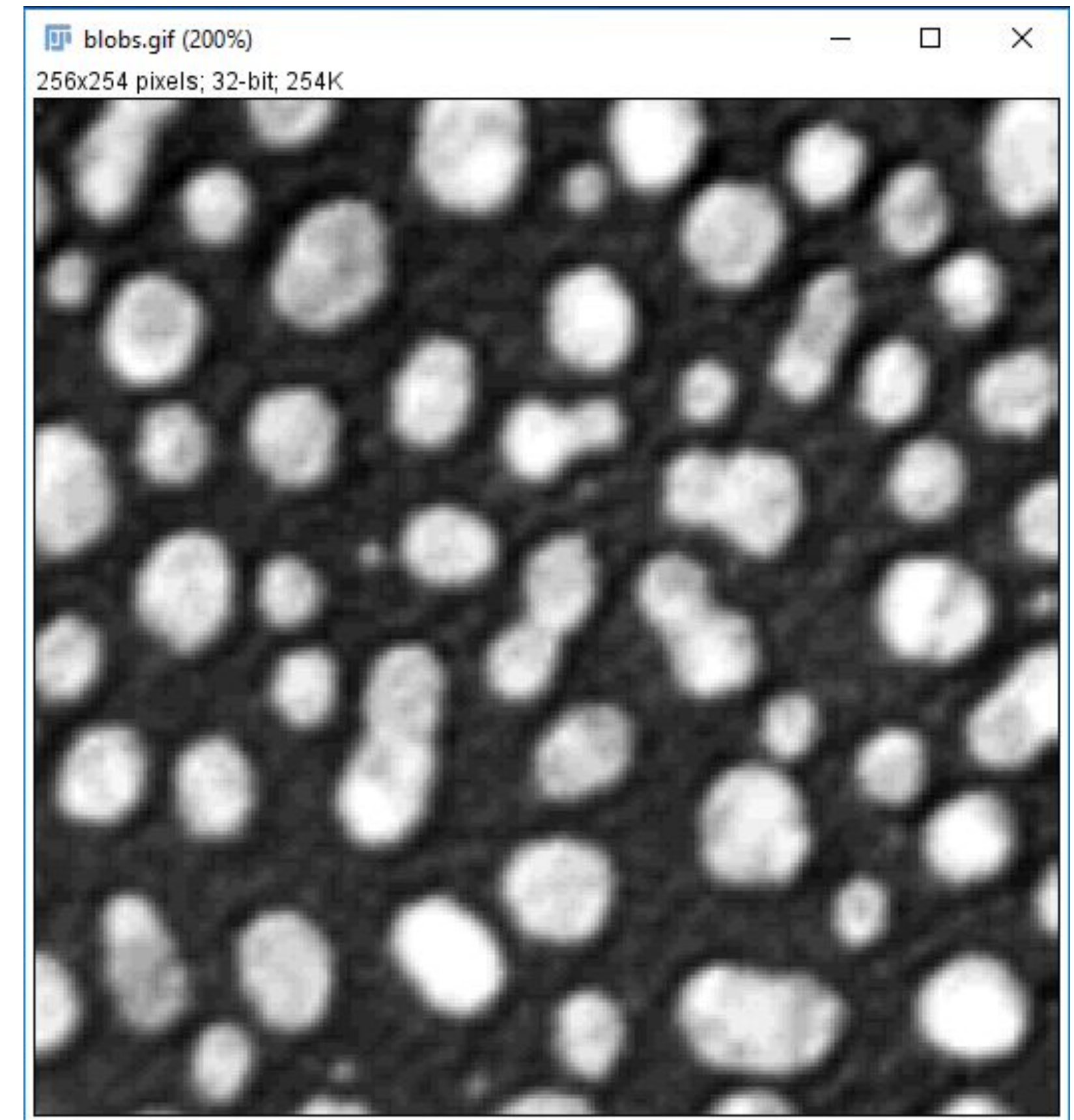
5x5

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |

[illegible]

# Convolution Example: Edge Detection

1. Open **Blobs** and transform it to 32-bit.  
Pay attention to the LUT.
2. Duplicate the image twice
3. On one duplicated image use the following filter:  
$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$
4. On the other one, use:  
$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$



# Convolution Example: Edge Detection

$$Result = \sqrt{Image_A^2 + Image_B^2}$$

5. Calculate the square of the two images:  
**Process**→**Math**→**Square** for each image
6. Calculate the sum of the processed images.  
**Process**→**Image Calculator**...
7. Merge the result with the original image:  
**Image**→**Color**→**Merge Channels**...
8. Adjust the Brightness for optimal visualization.



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